

Self-Compressing Vision Tower for Efficient Dense Prediction Tasks

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Date : August 15, 2025

URL :  <https://github.com/adishourya/SelfcompressingDepthWiseAttn>



1. Introduction

High-resolution dense prediction remains a critical task in computer vision. The effectiveness of a vision model is often evaluated by its performance both in accuracy and latency on large-scale dense prediction benchmarks.

Low-compute vision models enable real-time applications such as autonomous drones, on-device image processing, and computational photography. In domains like sports or wildlife photography, throughput can be just as important as accuracy. However, most state-of-the-art vision models rely on deep, sequential architectures, which are unsuitable for

deployment on consumer-grade devices with limited computational resources. In addition to their role in standalone vision tasks, vision towers that perform well on dense prediction problems are also considered as candidates for the visual encoders in vision-language models specially in small Vision Language Models [1].

Recent research suggest that the deeper layers of models may contribute less to overall accuracy than earlier layers, which indicates over-parameterization of networks and challenges the need for high computational cost in deeper layers [2], [3]. For instance, MiniMax-01 [4] employs full softmax attention ($O(N^2)$) only in few initial layers, while adopting

more compute efficient alternatives such as linear attention ($O(N)$) in the deeper layers to reduce total compute requirements.

Inspired by these insights, we propose a fully prunable and quantizable vision model as a backbone where all major learnable components such as convolution and attention weights and linear layers can be compressed and pruned to improve model storage efficiency and improve accuracy during low precision inference for dense prediction tasks.

A key hypothesis we examine in this work is the need for uniform allocation of model precision and compute cost across model depth. In a non self-compressing models [5], [6], weight precision and compute budgets are assigned identically to all layers regardless of their relative contribution to final performance. This uniform treatment can result in overparameterization and inefficient use of compute resources. We extensively evaluate our model on image classification, segmentation and optical flow problems. Our model matches accuracy over existing state of the art method. and present the following key contribution:

1. **Unified Compression Strategy:** We propose a unified framework that to quantize and compress the weights of modules proportionally to their computational cost, enabling consistent compression across all major learnable components Convolutional, Attention Weights, and Linear Layers via pruning and quantization.
2. **Efficient Deployment :** Our compressed model sustains accuracy levels comparable to the FP32 baseline during FP8 inference, while significantly improving throughput during inference.

2. Related Study

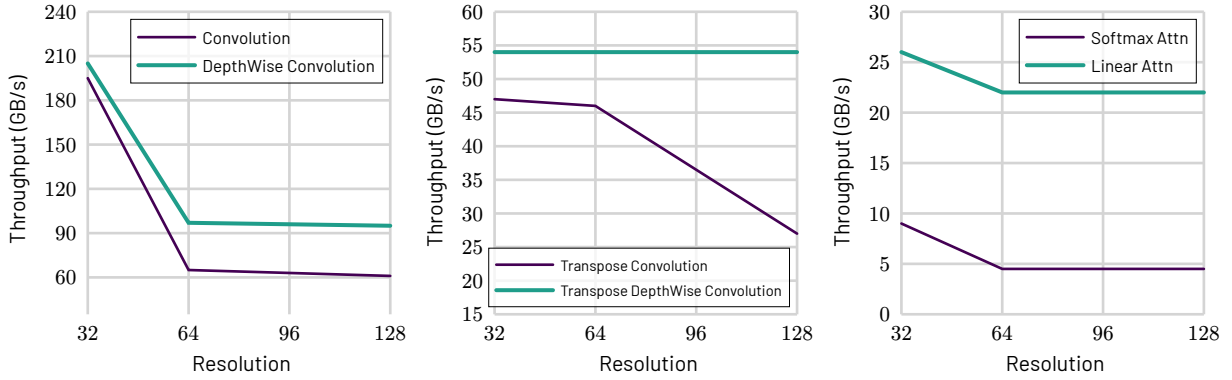
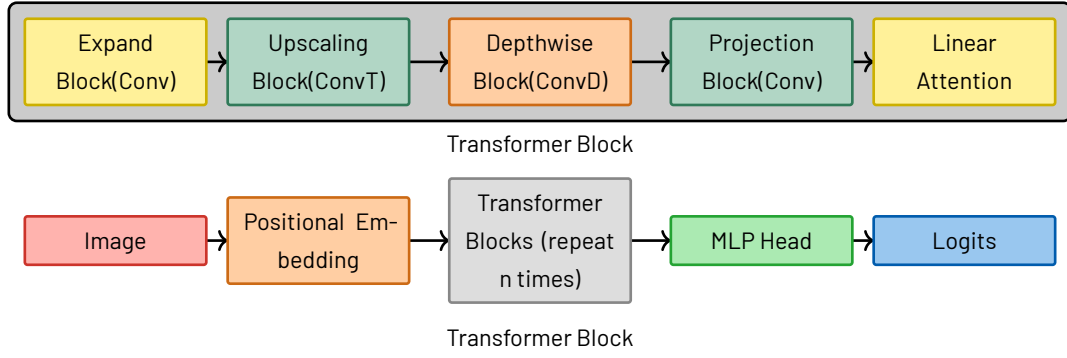
Our methodology builds upon recent advancements in efficient vision towers and quantization approaches, drawing insights from prior work. Below, we summarize key contributions from related works that have informed our approach.

1. **EfficientViT** [7] carefully selects computational modules to maximize accuracy while maintaining

high throughput. The architecture strategically offloads local inductive bias computation to light-weight convolution kernels, avoiding the computationally expensive full softmax attention. For example, it employs grouped depthwise convolutions to capture rich local structure at low cost, while using linear attention to obtain global receptive fields.

This careful allocation of operations results in significant speedups while maintaining state-of-the-art accuracy across dense prediction tasks. However, EfficientViT focuses on module selection and efficient operations without incorporating compression strategies, which could further improve both throughput and accuracy while reducing model storage cost.

2. **LayerDrop** [8] applies structured, layer-wise dropout in Transformer networks. During training, entire layers are randomly skipped either stochastically or through learned skipping which regularizes the network and makes it robust to layer removal. At inference, this enables extraction of arbitrary-depth sub-networks from a single model without retraining, providing scalability across compute budgets. While effective for pruning at the layer level, LayerDrop does not perform fine-grained compression such as bit-depth quantization or weight-level pruning.
3. **DiffQ** [9] presents a differentiable quantization framework that learns both weights and their bit depths without gradient approximations. To emulate the effect of quantization during training, it injects independent pseudo-quantization noise, enabling end-to-end optimization of precision at the level of individual weights or small groups, governed by a single accuracy-size trade-off hyperparameter. This design achieves competitive compression across vision, language, and audio tasks.
4. **Self-Compressing Networks** [10] remove the need for noise emulation by directly learning bit depths as trainable parameters, which can be reduced to zero to prune weights or entire channels. Unlike DiffQ's coalesced memory grouping, they adopt coarser mode-level (channel) grouping for



greater stability and reduced forgetting. They also treat number format parameters (scale, exponent, bit depth) as optimizable network parameters, allowing complete channel elimination without specialized hardware, and rely on the Straight-Through Estimator instead of pseudo-noise for gradient flow.

3. Method

We design our model by adapting ideas from recent vision models [7] and [11]. Our architecture follows a decoder-only transformer structure. The input image is first projected into a latent sequence of embeddings, where each embedding dimension corresponds to a learnable convolution kernel. Each transformer block is then implemented as a sequential composition of convolutional and transpose convolution operations [12], consisting of channel expansion (convolution), learned upscaling (grouped transposed convolution), depthwise convolution, and channel projection (convolution). After these local operations, a linear attention module is applied to capture global receptive fields as in Figure 1.

We quantize all matrices associated to the compute modules as the primary method to quantize and prune modes of weights. Specifically, the penalty of operations is designed to be proportional to the size of the parameter matrices and inversely proportional to the measured throughput on target device for each module. This penalty is incorporated into the training objective as a hinge-style regularizer, resulting in the overall loss function where L_o is the original loss of the model and γ acts as the compression factor over the normalizing constant C as in Equation 1.

$$\text{Loss} = L_o + \frac{\gamma}{C} \sum_{k=1}^{\text{all modules}} \alpha_k \text{size}_k \quad (1)$$

For, our experiments use quantization as used in [10] to quantize weights in the forward pass and use symmetric differentiable number Q8A format [13] given in Equation 2. The bit-depth b and the scaling factor s is shared per mode of parameters in W .

$$Q(W, b, s)_k = 2^e \lfloor \text{clip}\left(\frac{W}{2^s}, -2^{b-1}, 2^{b-1} - 1\right) \rfloor \quad (2)$$

The $\lfloor \cdot \rfloor$ acts as a straight through estimator [14] for the rounding function which returns the identity of upstream gradient during backward pass.

In the following sections, we derive the penalty cost for modules in our transformer block.

1. **Convolution:** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.
2. **Transposed Convolution:** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aequae doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguere possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et.
3. **Linear Attention:** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tem-

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3.1. Experiments

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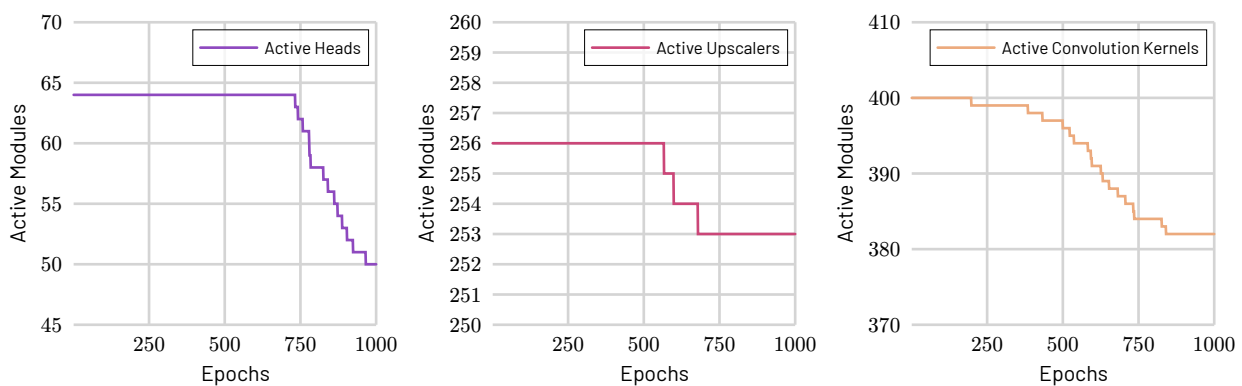


Figure 3: Module Pruning

hinc inimicus. Sed iure Mucius. Ego autem mirari satis non queo unde hoc sit tam insolens domesticarum rerum fastidium. Non est omnino hic docendi locus; sed ita prorsus existimo, neque eum Torquatum, qui hoc primus cognomen invenerit, aut torquem illum hosti detraxisse, ut aliquam ex eo est consecutus? – Laudem et caritatem, quae sunt vitae sine metu de-gendae praesidia firmissima. – Filium morte multavit. – Si sine causa, nollem me ab eo delectari, quod ista Platonis, Aristoteli, Theophrasti orationis ornamenta neglexerit. Nam illud quidem physici, credere aliquid esse minimum, quod profecto numquam putavisset, si a Polyaeno, familiari suo, geometrica discere maluisset quam illum etiam ipsum dedocere. Sol Democrito magnus videtur, quippe homini erudito in geometriaque perfecto, huic pedalis fortasse; tantum enim esse omnino in nostris poetis aut inertissimae segnitiae est aut fastidii delicatissimi. Mihi quidem videtur, inermis ac nudus est. Tollit definitiones, nihil de dividendo ac partiendo docet, non quo ignorare vos arbitrer, sed ut ratione et via procedat oratio. Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti

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3.2. Ablation Studies

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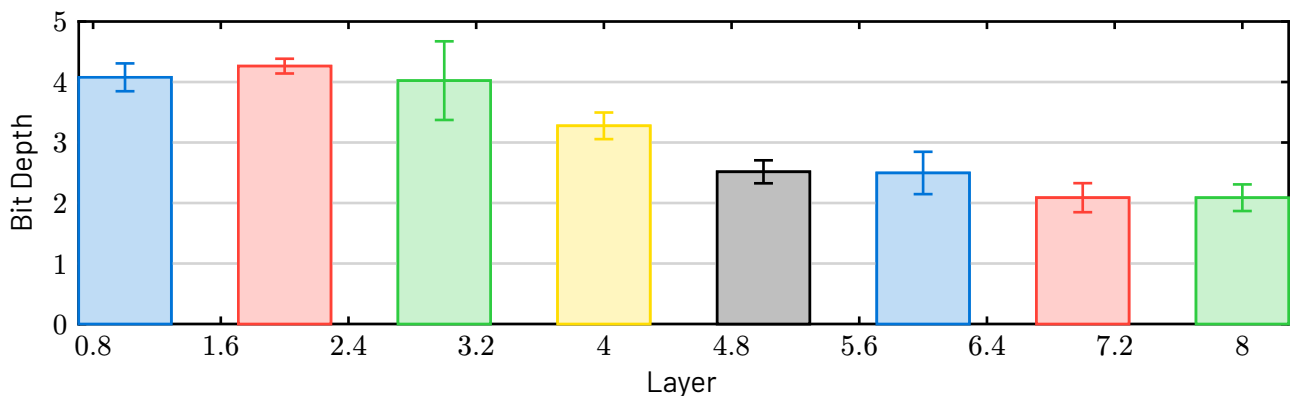


Figure 4: Pseudo Model Bit Depth Across Layers

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4. Evaluation

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Model	Size	Top1	Top5	FLOPS	Throughput H100 (Images/sec)
EfficientViT	0.98M	-	-	-	-
LayerDrop	1.12M	-	-	-	-
Ours	0.88M	-	-	-	-
EfficientViT	2.41M	-	-	-	-
LayerDrop	2.50M	-	-	-	-
Ours	1.98M	-	-	-	-

Table 1: cifar100

Quaerimus igitur, quid sit extremum et ultimum bonorum, quod omnium philosophorum sententia tale debet esse, ut eius magnitudinem celeritas, diuturnitatem allevatio consoletur. Ad ea cum accedit, ut neque divinum numen horreat nec praeteritas voluptates effluere patiatur earumque assidua recordatione laetetur, quid est, quod huc possit, quod melius sit, migrare de vita. His rebus instructus semper est in voluptate esse aut in armatum hostem impetum fecisse aut in poetis evolvendis, ut ego et Triarius te hortatore facimus, consumeret, in quibus hoc primum est in quo admirer, cur in gravissimis rebus non delectet eos sermo patrius, cum idem fabellas Latinas ad verbum e Graecis expressas non inviti

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Model	Size	Top1	Top5	FLOPS	Throughput H100 (Images/sec)
EfficientViT	0.98M	-	-	-	-
LayerDrop	1.12M	-	-	-	-
Ours	0.88M	-	-	-	-
EfficientViT	2.41M	-	-	-	-
LayerDrop	2.50M	-	-	-	-
Ours	2.13M	-	-	-	-

Table 2: country211

Model	Size	MIoU	FLOPS	Throughput H100 (Images/sec)
EfficientViT	2.41M	-	-	-
LayerDrop	2.50M	-	-	-
Ours	2.13M	-	-	-

Table 3: ade

Model	Size	EPE	FLOPS	Throughput H100 (Images/sec)
EfficientViT	2.41M	-	-	-
LayerDrop	2.10M	-	-	-
Ours	2.13M	-	-	-

Table 4: hd1k

5. Discussion

6. Conclusion

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A. Appendix

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B. Checklist /

B.A. Report Checklist

This checklist was created by Chang Sun for Bachelor's and Master's students at the Department of Advanced Computing Sciences, Maastricht University. It supports writing a complete, clear, and scientifically sound thesis. Students are encouraged to share this checklist with supervisors and highlight the items they have addressed.

B.A.A. Introduction

1. **Problem Statement:** Clearly define the key problem your thesis addresses. [research_question](#) 
2. **Significance:** Explain why this problem is important and urgent. [significance](#) 
3. **Research Challenges:** Identify why the problem has not been solved yet. 
4. **Brief Description of This Work:** Summarize your approach and how it differs from existing methods. [desc](#) 
5. **(Optional) Illustrative Diagram:** Include a simple, easy-to-understand diagram. 
6. **Key Contributions:** List and highlight the innovative contributions. [contribution](#) 
7. **Thesis Structure:** Briefly describe what each chapter covers. 

B.A.B. Related Work

1. Summarize state-of-the-art methods relevant to your thesis.
2. Discuss strengths and limitations.
3. Identify gaps in current work.
4. Explain how your thesis addresses these gaps.

B.A.C. Methodology

1. **Approach Overview:** High-level view with a well-labeled diagram 
2. **Components:** Explain each component in separate subsections.
3. **Implementation Details:**
 1. Method implementation
 2. Architecture choices, algorithms, and software
 3. Code must be open-source and documented on GitHub.

B.A.D. Datasets

1. **Data Sources & Collection:**

1. What, where, when, and how data was collected 
 2. Flowchart: source → filtering → final dataset 
 3. Inclusion/exclusion criteria and counts
 4. Provide access details in footnotes or appendix
2. **Dataset Description:**
1. Size, distributions, missing values
 2. Preprocessing steps

B.A.E. Experiments

1. **Design:**
 1. Dataset splits (training/test/validation)
 2. Which experiments are run, which parameters tested
2. **Settings:**
 1. Hardware (GPU/CPU model)
 2. Epochs, batch size, learning rate, etc.
3. **(Include figures/flowcharts when design is complex.)**

B.A.F. Evaluation Methodology

1. Describe performance evaluation strategy.
2. List and define metrics (e.g., accuracy, precision).
3. Include equations and implementation notes if possible.

B.A.G. Results

1. Use tables and figures to report results clearly.
2. Highlight key numbers readers should focus on.
3. **Statistical Significance:**
 1. Show deviations, error bars, or other significance metrics.
4. Objectively compare models, parameters—avoid interpretations here.

B.A.H. Discussion

1. Interpret findings and patterns.
2. Reflect on unexpected trends.
3. **Limitations:** What didn't work, assumptions, robustness.
4. **Future Work:** Propose directions for improvement.
5. **(Optional)** Discuss scientific and societal impacts.

B.A.I. Conclusion

1. Concise summary of your thesis.
2. Highlight discoveries and findings.
3. Share lessons learned.
4. **(If not in Discussion)** mention future work directions.

B.A.J. References

1. All references must include DOI or persistent IDs.
2. Prefer peer-reviewed papers. Preprints (e.g., arXiv) are allowed but disclose their proportion.

B.A.K. Chapter Organization

Sections 4–8 may be structured in one of the following ways:

1. Separate each as an individual chapter.
2. Combine:

1. **Datasets + Experiments + Evaluation** in one chapter;
2. **Results + Discussion** in another.
3. Or:
 1. **Dataset** in one chapter;
 2. **Experiments, Evaluation, Results** in another;
 3. **Discussion** separately.
4. Other logical groupings are allowed if all parts are clearly covered.

B.B. Defense Presentation Checklist

1. What is the problem?
2. Why is it significant?
3. Related work
4. Research questions
5. Datasets, study design, and experiments
6. Results and how they answer the research questions
7. List of key contributions
8. Future work directions